



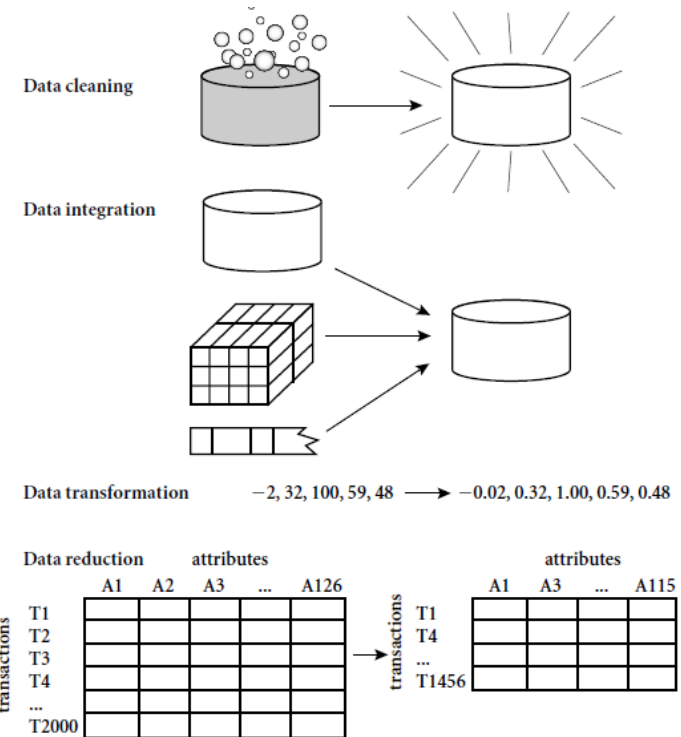
Applied Data Mining

**— Week 3—
Spring 2026**

Dr. Hashim Safdar

Outline

- Data Preprocessing: Definition, Goal
- Why do we preprocess the data?
- Major Tasks in Data Preprocessing
 - Data Cleaning
 - Data Integration
 - Data Reduction
 - Data Transformation



Data Integration



- **Data integration:**
 - Combines data from multiple sources into a **coherent** store
- Schema integration: e.g., $A.cust-id \equiv B.cust-\#$
 - Integrate metadata from different sources
- **Entity identification problem:**
 - Identify|real-world entities from multiple data sources, e.g., Bill Clinton = William Clinton
- Detecting and resolving data value conflicts
 - For the same real-world entity, attribute values from different sources are different
 - Possible reasons: different representations, different scales, e.g., metric vs. British units

Metadata= Structure of Data

Handling Redundancy in Data Integration

- Redundant data often occur when we integrate multiple databases
 - *Object identification*: The same attribute or object may have different names in different databases
 - *Derivable data*: One attribute may be a “derived” attribute in another table, e.g., annual revenue
- Redundant attributes may be able to be detected by *correlation analysis* and *covariance analysis*
- Careful integration of the data from multiple sources may help reduce/avoid redundancies and inconsistencies and improve mining speed and quality

Handling Redundancy in Data Integration

- Some redundancies can be detected by **correlation analysis**.
- Given two attributes, such analysis can measure how strongly one attribute implies the other, based on the available data.
- For nominal data, we use the (*chi-square*) test. Example:
Gender, Marital Status
- For numeric attributes, we can use the *correlation coefficient* and *covariance*, both of which access how one attribute's values vary from those of another.

Correlation Analysis (Nominal Data)

χ^2 (chi-square) test

discovered by a χ^2 (chi-square) test. Suppose A has c distinct values, namely $a_1, a_2, \dots, a_c$. B has r distinct values, namely $b_1, b_2, \dots, b_r$. The data tuples described by A and B can be shown as a contingency table, with the c values of A making up the columns and the r values of B making up the rows. Let (A_i, B_j) denote the joint event that attribute A takes on value a_i and attribute B takes on value b_j , that is, where $(A = a_i, B = b_j)$. Each and every possible (A_i, B_j) joint event has its own cell (or slot) in the table. The χ^2 value (also known as the *Pearson χ^2 statistic*) is computed as

$$\chi^2 = \sum_{i=1}^c \sum_{j=1}^r \frac{(o_{ij} - e_{ij})^2}{e_{ij}}, \quad (3.1)$$

where o_{ij} is the *observed frequency* (i.e., actual count) of the joint event (A_i, B_j) and e_{ij} is the *expected frequency* of (A_i, B_j) , which can be computed as

Correlation Analysis (Nominal Data)

χ^2 (chi-square) test

$$e_{ij} = \frac{\text{count}(A = a_i) \times \text{count}(B = b_j)}{n}, \quad (3.2)$$

where n is the number of data tuples, $\text{count}(A = a_i)$ is the number of tuples having value a_i for A , and $\text{count}(B = b_j)$ is the number of tuples having value b_j for B . The sum in Eq. (3.1) is computed over all of the $r \times c$ cells. Note that the cells that contribute the most to the χ^2 value are those for which the actual count is very different from that expected.

- The χ^2 tests the hypothesis that A and B are *independent*, that is, there is no correlation between them.

$$\chi^2 = \sum \frac{(\text{Observed} - \text{Expected})^2}{\text{Expected}}$$

Correlation Analysis (Nominal Data)

Example

Correlation analysis of nominal attributes using χ^2 . Suppose that a group of 1500 people was surveyed. The gender of each person was noted. Each person was polled as to whether his or her preferred type of reading material was fiction or nonfiction. Thus, we have two attributes, *gender* and *preferred_reading*. The observed frequency (or count) of each possible joint event is summarized in the contingency table shown in Table 3.1, where the numbers in parentheses are the expected frequencies. The expected frequencies are calculated based on the data distribution for both attributes using Eq. (3.2).

Using Eq. (3.2), we can verify the expected frequencies for each cell. For example, the expected frequency for the cell (*male*, *fiction*) is

$$e_{11} = \frac{\text{count}(\text{male}) \times \text{count}(\text{fiction})}{n} = \frac{300 \times 450}{1500} = 90,$$

Chi-Square Calculation: An Example

	Male	Female	Total (row)
fiction	250(90)	200(360)	450
Non- fiction	50(210)	1000(840)	1050
Total (col.)	300	1200	1500

- χ^2 (chi-square) calculation:

$$\chi^2 = \frac{(250 - 90)^2}{90} + \frac{(50 - 210)^2}{210} + \frac{(200 - 360)^2}{360} + \frac{(1000 - 840)^2}{840} = 507.93$$

- It shows that gender and preferred_reading are strongly correlated in the group.

Chi-Square Calculation: An Example

- The χ^2 distribution is an asymmetric distribution that has a minimum value of 0, but no maximum value.
- The curve reaches a peak to the right of 0, and then gradually declines in height, the larger the χ^2 value is.

Correlation Analysis (Numeric Data)

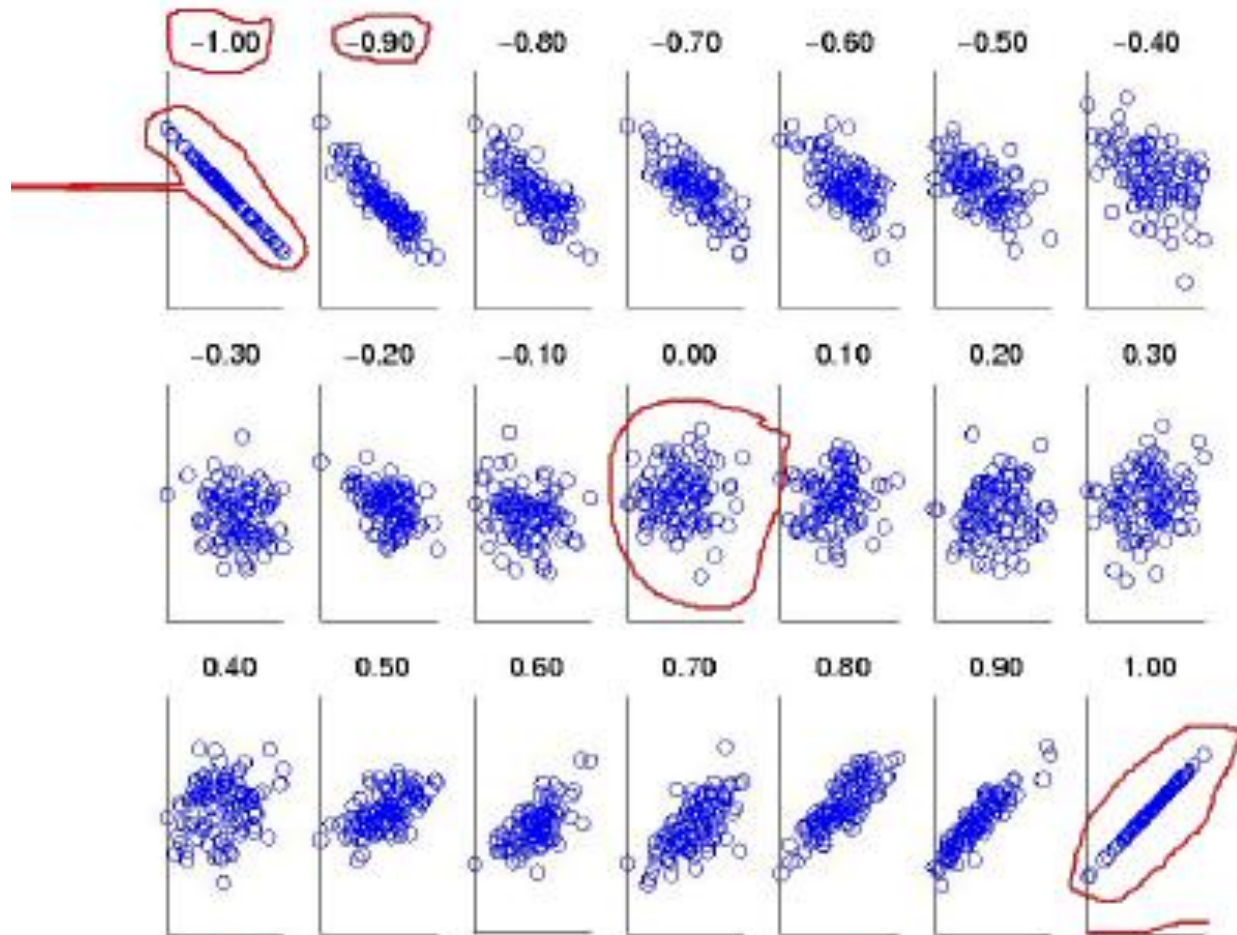
- Correlation coefficient (also called Pearson's correlation coefficient)

$$r_{A,B} = \frac{\sum_{i=1}^n (a_i - \bar{A})(b_i - \bar{B})}{(n-1)\sigma_A\sigma_B} = \frac{\sum_{i=1}^n (a_i b_i) - n\bar{A}\bar{B}}{(n-1)\sigma_A\sigma_B}$$

where n is the number of tuples, $\bar{A}$ and $\bar{B}$ are the respective means of A and B , σ_A and σ_B are the respective standard deviation of A and B , and $\sum(a_i b_i)$ is the sum of the AB cross-product.

- If $r_{A,B} > 0$, A and B are positively correlated (A 's values increase as B 's). The higher, the stronger the correlation.
- $r_{A,B} = 0$: independent; $r_{AB} < 0$: negatively correlated

Visually Evaluating Correlation



Scatter plots
showing the
similarity from
-1 to 1.

Correlation measures the linear relationship between objects

CA (Numeric Data)...

Example1:

Does financial status affect health? The observed correlation between household income and the prevalence of coronary artery disease is $r = -0.717$.

- So wealthier you are, the lower your risk of having a heart attack (strong negative correlation).